

Quick Prototype Specification – PM2.5 & Breathing Rate Monitor (V1.0)

Objective

Develop a low-cost proof-of-concept device capable of:

- Measuring ambient PM2.5 concentration
- Estimating breathing rate
- Displaying and transmitting data via Bluetooth
- Running on a rechargeable battery

This is an engineering prototype intended for demonstration and validation, not a medical device.

Hardware Components

Component	Suggested Part	Notes
Microcontroller	ESP32-C3 Super Mini	Wi-Fi + BLE, USB programming
PM Sensor	Plantower PMS5003	UART interface, PM1.0/PM2.5/PM10
Heart/Breath Sensor	MAX30102	I ² C interface, breathing rate derived from PPG
Display	0.96" OLED (SSD1306)	I ² C
Battery	3.7 V Li-ion/LiPo (1000–2000 mAh)	Rechargeable
Charging IC	TP4056 USB-C Module	Battery charging & protection

Voltage Regulator	3.3 V LDO	If required
Push Button	Reset/User Button	Optional
Status LED	Green/Red	Device status

Electrical Interfaces

Device	Interface	ESP32 Pins (Suggested)
MAX30102	I ² C	SDA, SCL
OLED	I ² C	Shared SDA, SCL
PMS5003	UART	RX, TX
Battery	Power	3.7 V
USB-C	Programming & Charging	Native USB

PCB Requirements

PCB

- 2-layer FR4 PCB
- Thickness: 1.6 mm
- Copper: 1 oz
- PCB Size: Approximately 70 × 50 mm
- Mounting holes: 4 × M3
- JST battery connector
- Header pins for PMS5003 (or JST connector)
- USB-C programming access
- Test pads for power and UART

Component Placement

- MAX30102 positioned near PCB edge for unobstructed finger placement.

- PMS5003 connected via cable and mounted externally to allow unrestricted airflow.
 - OLED on top side for visibility.
 - Battery mounted beneath PCB using adhesive.
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Firmware Functions

1. Initialize all sensors.
 2. Read PM2.5 data every 5 seconds.
 3. Read PPG continuously from MAX30102.
 4. Estimate breathing rate (breaths/min) using PPG waveform analysis.
 5. Display:
 - PM2.5
 - Heart Rate
 - Estimated Respiratory Rate
 6. Broadcast data over Bluetooth Low Energy (BLE).
 7. Enter low-power mode between readings where appropriate.
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Mechanical Assembly

- 3D-printed enclosure with ventilation openings for the PMS5003.
 - Finger opening or clip for MAX30102 measurement.
 - Accessible USB-C charging port.
 - Battery secured with foam tape.
 - Easy access for PCB debugging and firmware updates.
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Deliverables

PCB Design

- KiCad project files
- Schematic
- PCB layout
- Gerber files
- Bill of Materials (BOM)

- Pick-and-Place (Centroid) files

Manufacturing

- Fabricate 5 prototype PCBs
- Assemble all SMT and through-hole components
- Functional power-on test

Firmware

- ESP32 source code
 - Sensor drivers
 - BLE implementation
 - Documentation for flashing firmware
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Prototype Success Criteria

- Device powers on reliably.
- PM2.5 readings update every 5 seconds.
- Heart rate and estimated breathing rate are displayed.
- BLE transmits sensor data to a smartphone or laptop.
- Battery operation for at least 4 hours.
- Stable operation for a minimum of 30 minutes without resets.

Target Timeline: 2–3 weeks (PCB design, fabrication, assembly, and firmware integration).

This specification is intentionally scoped for a rapid prototype so a hardware team can move directly into schematic capture, PCB layout, fabrication, assembly, and initial firmware development without unnecessary complexity.